Problem 2. Dew Point Calculations

We begin by stating the composition of Dry air. The data are taken from Mackenzie, F.T. and J.A. Mackenzie (1995) Our Changing Planet, Prentice-Hall, Upper Saddle River, NJ pp. 288-307

N_2	78.084	% mol
O_2	20.947	% mol
Argon	0.934	% mol
CO_2	0.035	% mol

Trace amounts are added into Argon. Notice that the composition of CO_2 might be higher today.
We now must calculate the composition of the humid air. From the definition of Relative Humidity

$$RH = \frac{P_{H_2O}}{P_{H_2O}^{sat}} \Rightarrow RH = \frac{y_{H_2O} P}{P_{H_2O}^{sat}}$$

At $T = 104\ °F$ and $P = 29.39$ in Hg and $RH = 0.72$ the $P_{H_2O}^{sat}$ from the Steam Properties of the CHE 301 Calculator we obtain $P_{H_2O}^{sat} = 1.07$ psia $= 2.178542$ in Hg and therefore

$$y_{H_2O} = \frac{(RH)\, P_{H_2O}^{sat}}{P} = \frac{(0.72)(2.178542)}{29.39} \Rightarrow \boxed{y_{H_2O} = 0.0534}$$

The composition of all the components in the humid air is then $y_i = (1 - y_{H_2O})\, y_i^{DRY}$, $i = N_2, O_2, Ar, CO_2$

The composition of the humid air obtained therefore is

1.	$N_2 = 0.73917$	$T_c = -146.95\ °C$: Supercritical
2.	$O_2 = 0.19829$	$T_c = -118.55\ °C$: Supercritical
3.	$Ar = 0.00884$	$T_c = -122.35\ °C$: Supercritical
4.	$CO_2 = 0.00033$	$T_c = 31.05$
5.	$H_2O = 0.05337$	$T_c = 374.15$

Assuming that the air behaves as ideal gas and that the composition of the gases in the liquid can be calculated by Raoult's law and Henry's law for the supercritical components as shown in the table above, the dew point equation becomes

$$\sum \frac{z_i}{K_i} = 1 \Rightarrow \frac{z_1 P}{H_1} + \frac{z_2 P}{H_2} + \frac{z_3 P}{H_3} + \frac{z_4 P}{H_4} + \frac{z_5 P}{P^{sat}_{H_2O}} = 1$$

where for CO_2 we have also used Henry's law because the critical temperature of CO_2 is close to the dew point.
The values of H_1, H_2, H_3 and H_4 are $H_1 = 87{,}650$ bar, $H_2 = 44{,}380$ bar $H_3 = 40{,}000$ bar and $H_4 = 1{,}670$ bar. We use the calculator to get the dew point by using the following steps.

Step 1. Enter Components and molar composition

Step 2. Enter Pressure and a guess for dew Temperature $T^o = 30°C$

Step 3. Calculate P^{sat} for water and enter Henry's constants for all other components

Step 4. Calculate the K-values

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Step 5. Set the value of $\alpha = 1$

Step 6. Solve for the DewPoint temperature using Goalseek

For Goalseek Set Cell: Rachford-Rice Equation

To Value: 0.0

By changing cell: Dew Point Temperature

An image of the Excel Calculations with the Set cell in **blue** and the changing cell in **red** is shown below:

Fluid Conditions	Mixture	N2	O2	Ar	CO2	H2O
Molar Composition	1.0000	0.73917	0.19829	0.00884	0.00033	0.05337
Temperature, C	33.7690					
Pressure, bar	0.9953					
Saturated Vapor Pressure, bar		87650.00000	44380.00000	40000.00000	1670.00000	0.05312
Saturated Vapor Temperature, C						
K-Values		88067.47240	44591.37964	40190.51793	1677.95412	0.05337
New K-Values						
Vapor to Feed Molar Ratio [a=V/F]	1.0000					
The Rachford-Rice Equation	0.0000					
Discriminant of the Cubic EOS						